Ex. No.:5	AUTOMATED LEGAL DOCUMENT ANALYZER USING GPT-4	URK23CS1167
09.10.2026		

Aim:

To develop an automated legal document analyzer using GPT-4 that summarizes legal documents and extracts important information such as parties, key clauses, obligations, payment terms, termination conditions and important dates.

Problem Statement:

Design and develop an automated legal document analyzer using GPT-4. The application should accept legal contracts or legal text, analyze the content, generate a concise summary, identify important clauses and obligations, extract relevant information and present the results in a structured format.

Software Requirements:

Software	Version
Python	3.13.18
Ollama	0.6.2
VS Code	1.133.0
Pandas	3.0.5
Sentence Transformers	5.7.0
Flask	3.1.3
LLM Model	Llama 3.2 (3B) via Ollama

Input Parameters:

Parameter	Description
Legal Document	Contract or legal text – uploaded .txt/.pdf/.docx file or pasted text. Example: employment_agreement.txt
Document Type	Auto-detect, Employment, NDA, Rental, Service, Sales, Loan or Other
Parties	Parties mentioned in the document (extracted automatically)
Document Length	Number of characters, words and approximate pages (calculated automatically)
Analysis Type	Complete, Summary, Clauses or Obligations
Output Format	Structured text report with headings and bullet points

System Architecture:

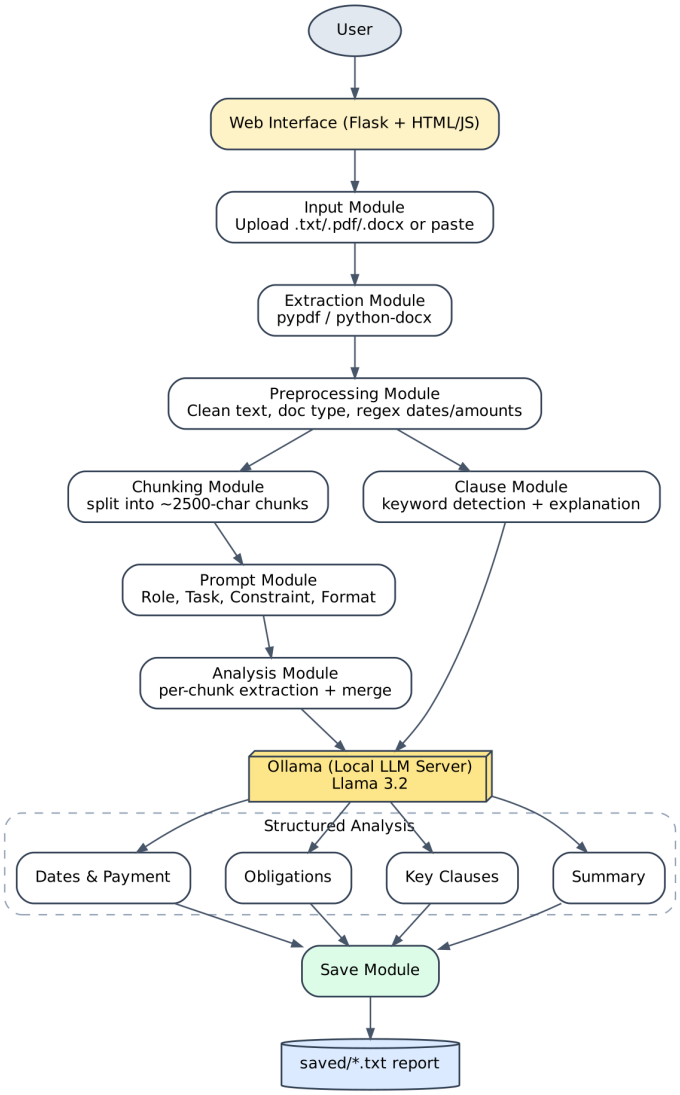


Fig. 5.1 System architecture

Module Design:

Module	Responsibility
Input Module	Accepts legal documents through file upload (.txt, .pdf, .docx), sample selection or manual text entry.
Extraction Module	Reads the uploaded document and extracts its text using pypdf and python-docx.
Preprocessing Module	Cleans the text, detects the document type and finds dates, amounts and durations using regular expressions.
Chunking Module	Splits long documents into chunks of about 2500 characters at paragraph boundaries.
Prompt Module	Creates a structured prompt with Role, Task, Information to extract, Constraint ('do not invent facts') and JSON Output Format.

Module	Responsibility
Analysis Module	Sends each chunk to the LLM, merges the extracted parties, obligations, payment, termination and dates, and generates an abstractive summary.
Clause Module	Identifies key clauses (confidentiality, payment, termination, liability, IP, dispute resolution) and explains each in simple words with importance.
Save Module	Saves the complete structured analysis report as a .txt file.

Implementation:

• GitHub Link - https://github.com/MONICK10/GPT4-Exp5-Legal-Document-Analyzer
• LinkedIn Post Link - <add LinkedIn post link in config.json>
• Run: `python app.py` → `http://localhost:5005`

Output Screenshots:

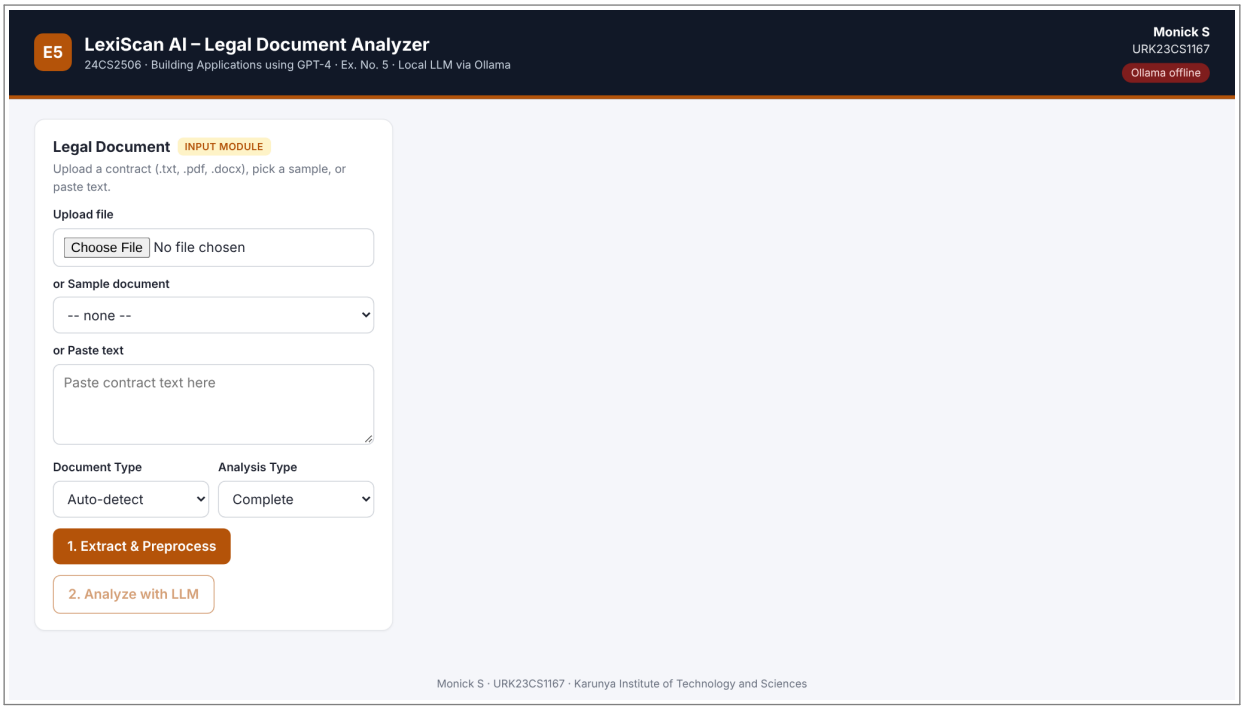


Fig. 5.2 Home Screen

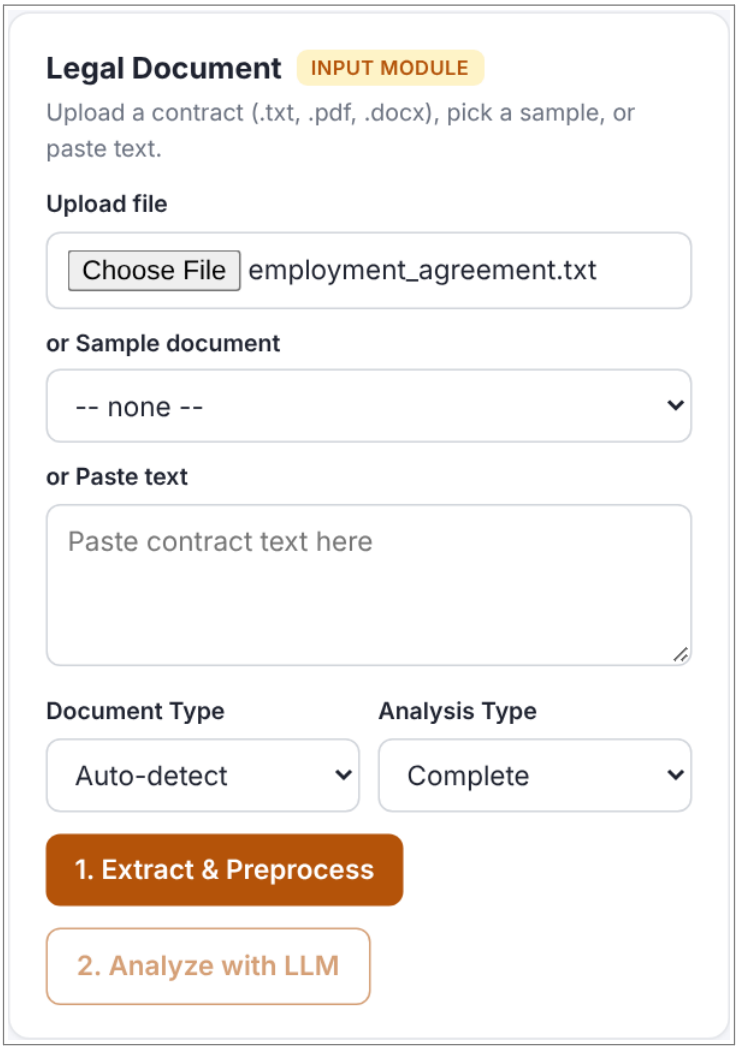


Fig. 5.3 Document Upload

[Draft] The AI output screenshots are added automatically when you run python tools/capture.py 5 on your laptop with Ollama.

Result:

Thus, an automated legal document analyzer using GPT-4 was successfully developed. The system accepts legal documents, processes the text, generates a concise summary, identifies important clauses and obligations, extracts relevant information and produces a structured analysis.